2026 BROCK AI DAY · POSTER · EXPLAINABLE AI / DEEP LEARNING

Are Language Models Converging on Shared Structures?

Mechanistic Interpretability, the Linear Representation Hypothesis, and the Platonic Representation Hypothesis

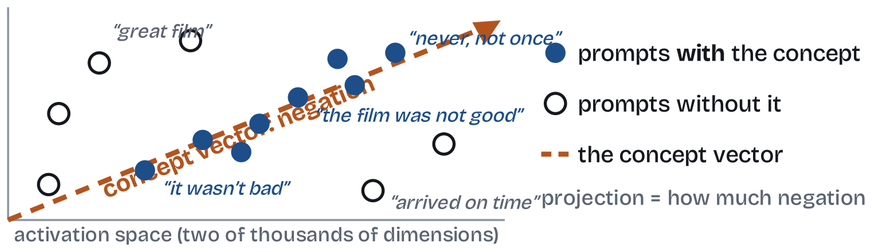
James R. A. Henry Computer Science (Undergraduate), Brock University jh13nt@brocku.ca · jamesrahenry@henrynet.ca github.com/jamesrahenry/ai-day-2026

1 Concepts are directions: the Linear Representation Hypothesis

Mechanistic interpretability reads a model's internal activations to find what it represents and how, rather than judging it by its outputs alone. It needs the model's weights, so it is only possible with open models.

A concept such as negation or sentiment is a *vector* in activation space (Park, Choe & Veitch, 2024). Prompts carrying the concept cluster along it, further out the stronger; projecting any activation onto it reads the concept off as a number. It bends where concepts share dimensions (superposition; Elhage et al., 2022) and from one layer to the next.

WHERE THE VECTOR LIVES The **residual stream**: one running vector per token that every layer's attention and MLP blocks read from and add their output back into. We read it at the *last token* of a passage, after the model has seen all of it, at any layer.



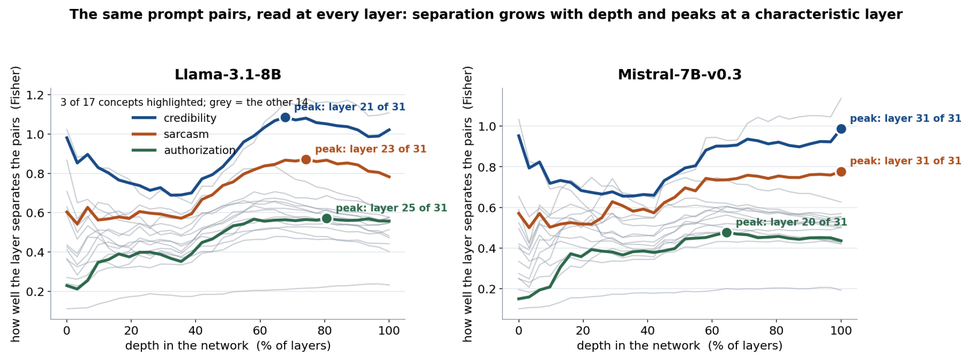
2 Extracting a concept vector from contrastive pairs

Pairs. For each concept, write passages on the same topic in two versions, one carrying the concept and one not. Here: 17 concepts, about 250 pairs each, written by 14 different LLMs. Run both versions through the model and record the residual stream at the final token of each passage, at every layer.

WITH: CREDIBILITY MARKERS
"Traditional vaccine development is a meticulous process, typically requiring 10–15 years..."

WITHOUT: SAME CLAIM, NO GROUNDING
"It's just common sense that a vaccine should take years to develop..."

Difference of means. At each layer, the mean activation of the *with* passages minus the mean of the *without* passages is that layer's concept vector (Zou et al., 2023; Rimsky et al., 2024). Score each layer by how well its vector separates the two sets. The peak layer is where the concept is most cleanly represented.



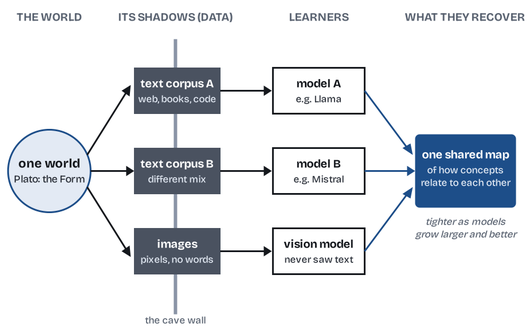
Separation at every layer for all 17 concepts in two models; credibility, sarcasm and authorization highlighted. Dots mark each concept's peak layer. Grammatical concepts such as negation and plurality peak early; sarcasm and authorization peak late.

3 Convergence: the Platonic Representation Hypothesis

Plato's cave: prisoners see only shadows cast by things they never see directly. The shadows change with the light; the things do not.

The hypothesis (Huh, Cheung, Wang & Isola, 2024): text, images and other data are shadows of one world, and models trained on different shadows, with different architectures and objectives, converge toward a shared model of that world, more so as they grow.

What converges is relations, not coordinates: which things are similar to which.
Prediction here: models that never shared a weight should agree on how concepts relate to each other. Section 4 tests it.



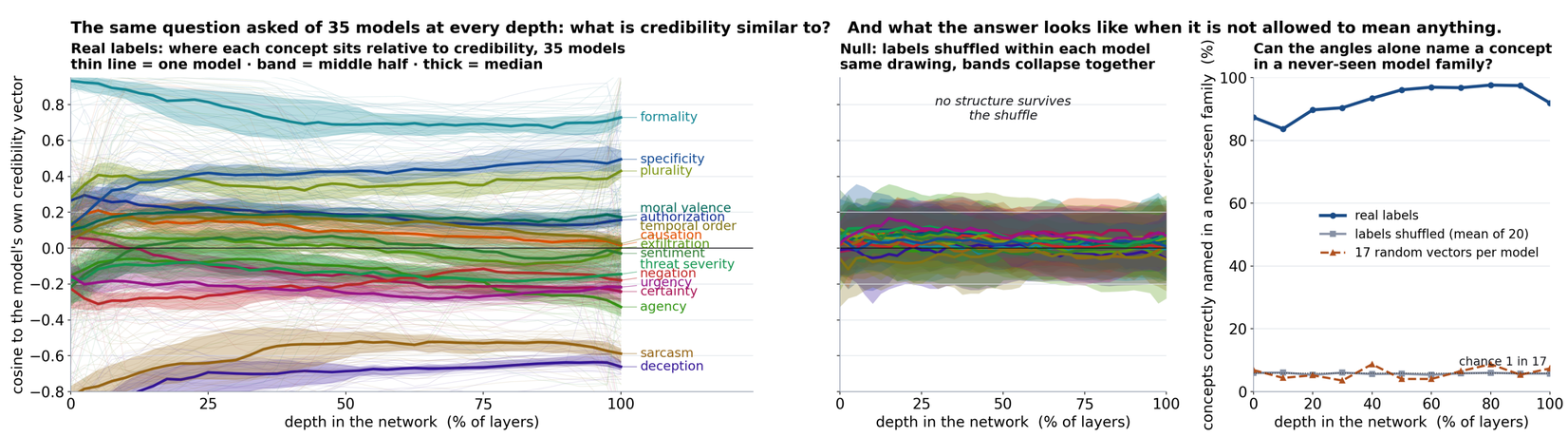
Different shadows, different learners, one structure recovered; the coincidence tightens with scale.

4 Our test: ask 35 models the same question

Each model's credibility vector lives in its own coordinate system, so two models' vectors cannot be laid on each other. What *can* be compared are the angles inside each model (relative representations: Moschella et al., 2023).

So ask 35 models, at every layer: **what is credibility similar to?** Formality sits closest in all of them, specificity next; sarcasm and deception sit far away. Wider bands mean looser agreement across models; all bands tighten from mid-stack on. The middle panel shows the shuffle.

The test that can fail: hold out a whole model family, describe each of its concept vectors only by its angles to the others. Mid-depth: 97% identified correctly. Shuffled labels or random vectors: 6%, which is chance.



What it might mean: the consolidation of meaning

Models built by different teams on different text arrive at the same map of how concepts relate. That is the distributional hypothesis made visible: meaning is the company a word keeps (Harris, Firth); a concept's value is its position among the others (Saussure). Whether such structure amounts to meaning is contested (Piantadosi & Hill vs. Bender & Koller). What is new is that the map is measurable, and shared.

Is this the structure of language itself, or, as Plato would have it, the structure of the world it describes?

github.com/jamesrahenry/ai-day-2026
Poster source, figure code, full reading list and sources.

THIS WORK

Henry (2026a). Concept Allocation Zones. arXiv:2605.24856.

Henry (2026b). Geometric Evolution Maps. arXiv:2605.25848.

Henry (2026c). CAZ Validation. Zenodo 10.5281/zenodo.22016917.

Data: huggingface.co/datasets/james-ra-henry/Rosetta-Activations